

Andreas Ziegler

Robotics & Computer Vision
Researcher/Engineer

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in [andreas-ziegler](#)



Summary

I am a broadly trained roboticist with a passion for application-driven robotics, computer vision, and machine learning research. I developed novel algorithms and implemented them to work on real robots. Having lived, studied, and worked in diverse environments—including Zurich, Basel, Lausanne, Tübingen, Paris, Shanghai, and Seoul/Incheon—I appreciate the opportunity to collaborate and exchange ideas with individuals from diverse backgrounds. Diverse perspectives foster creativity and innovation, enabling teams to navigate challenges and deliver impactful results.

I am committed to contributing to a collaborative work environment that embraces experimentation, celebrates learning from failures, and ensures mutual respect and openness. I believe such a culture empowers individuals to take ownership, enabling both personal and collective growth.

As a Postdoctoral Researcher, I seek to grow as an individual contributor while leveraging my leadership experience to foster team collaboration and drive impactful results. By transitioning from individual successes to driving collective achievements, I aspire to stimulate the field of robotics, computer vision, and machine learning.

Personal details

Nationality Swiss

Education

- 2021.06–2026.01 **PhD**, *University of Tübingen*, Tübingen, Germany
Thesis: Event-based computer vision for fast robot control
○ In collaboration with Sony AI contributing to project ACE.
○ PhD committee (supervisors): Guillermo Gallego, Andreas Geiger, Georg Martius, and Andreas Zell
- 2014.09–2018.04 **MSc ETH in EEIT**, *ETH Zürich*, Zürich, Switzerland
Specialized in: Robotics, Computer Vision and Machine Learning
Master Thesis: A Representation for Exploration that is Robust to State Estimate Drift
○ Examiner: Prof. Dr. Roland Siegwart and Prof. Dr. Davide Scaramuzza
○ Resulted in [13]
Semester Project 2: Map Fusion for Collaborative UAV SLAM
○ Examiner: Prof. Dr. Roland Siegwart and Prof. Dr. Margarita Chli
Semester Project 1: Robust object tracking in 3D by fusing ultra-wideband and vision
○ Examiner: Prof. Dr. Luc Van Gool and Prof. Dr. Otmar Hilliges
- 2009.09–2013.09 **BSc FHO in Electrical Engineering**, *University of Applied Science Eastern Switzerland (HSR)*, Rapperswil, Switzerland
Specialized in: Digital Signal and Image Processing, Embedded Systems and Software Engineering, and Mobile Communication
- 2011.09–2012.08 **Exchange year**, *Shanghai Jiao Tong University*, Shanghai, China
Courses taken: Chinese language, Electrical engineering and Computer Science

Publications

- [1] C. Caccavella, V. Fra, A. Ziegler, G. D'Angelo, and Y. Sandamirskaya, "Efficient semantic understanding from digital foveation," in *Computer Vision – ECCV 2026 Workshops*. Springer Nature Switzerland, 2026. DOI: [arXiv.2609.04088](https://arxiv.org/abs/2609.04088)
- [2] T. Gossard, F. Radovic, A. Ziegler, and A. Zell, "BlurBall: Joint ball and motion blur estimation for table tennis ball tracking," in *2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2026. [Online]. Available: <https://arxiv.org/abs/2509.18387>
- [3] T. Gossard, J. Schmalzl, A. Ziegler, and A. Zell, "Spin detection using racket bounce sounds in table tennis," in *2025 IEEE International Workshop on Sport, Technology and Research (STAR)*, IEEE, Oct. 2025. [Online]. Available: <https://arxiv.org/abs/2409.11760>
- [4] T. Gossard, A. Ziegler, and A. Zell, "TT3D: Table tennis 3D reconstruction," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2025, pp. 5821–5831. DOI: 10.1109/cvprw67362.2025.00581 [Online]. Available: <http://dx.doi.org/10.1109/CVPRW67362.2025.00581>
- [5] A. Ziegler, D. Joseph, T. Gossard, E. Moldovan, and A. Zell, "BiasBench: A reproducible benchmark for tuning the biases of event cameras," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2025, pp. 4994–5003. DOI: 10.1109/cvprw67362.2025.00493 [Online]. Available: <http://dx.doi.org/10.1109/CVPRW67362.2025.00493>
- [6] A. Ziegler, K. Vetter, T. Gossard, J. Tebbe, S. Otte, and A. Zell, "Detection of fast-moving objects with neuromorphic hardware," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2025, pp. 8709–8717. DOI: 10.1109/icra55743.2025.11127960 [Online]. Available: <http://dx.doi.org/10.1109/ICRA55743.2025.11127960>
- [7] T. Gossard, J. Krismer, A. Ziegler, J. Tebbe, and A. Zell, "Table tennis ball spin estimation with an event camera," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2024, pp. 3347–3356. DOI: 10.1109/cvprw63382.2024.00339 [Online]. Available: <http://dx.doi.org/10.1109/CVPRW63382.2024.00339>
- [8] T. Gossard, A. Ziegler, L. Kolmar, J. Tebbe, and A. Zell, "eWand: An extrinsic calibration framework for wide baseline frame-based and event-based camera systems," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2024, pp. 14534–14540. DOI: 10.1109/icra57147.2024.10610116 [Online]. Available: <http://dx.doi.org/10.1109/ICRA57147.2024.10610116>
- [9] T. Gossard, J. Tebbe, A. Ziegler, and A. Zell, "SpinDOE: A ball spin estimation method for table tennis robot," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, Oct. 2023. DOI: 10.1109/iros55552.2023.10342178 [Online]. Available: <http://dx.doi.org/10.1109/IROS55552.2023.10342178>
- [10] A. Ziegler, T. Gossard, K. Vetter, J. Tebbe, and A. Zell, "A multi-modal table tennis robot system," in *RoboLetics: Workshop on Robot Learning in Athletics @CoRL 2023*, 2023. DOI: 10.48550/arXiv.2310.19062 [Online]. Available: <https://arxiv.org/abs/2310.19062>
- [11] A. Ziegler, D. Teigland, J. Tebbe, T. Gossard, and A. Zell, "Real-time event simulation with frame-based cameras," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2023. DOI: 10.1109/icra48891.2023.10160654 [Online]. Available: <http://dx.doi.org/10.1109/ICRA48891.2023.10160654>

- [12] A. N. Horvath et al., “Focus on time: Dynamic imaging reveals stretch-dependent cell relaxation and nuclear deformation,” *Biophysical Journal*, vol. 120, no. 5, pp. 764–772, Mar. 2021, ISSN: 0006-3495. DOI: 10.1016/j.bpj.2021.01.020 [Online]. Available: <http://dx.doi.org/10.1016/j.bpj.2021.01.020>
- [13] T. Cieslewski, A. Ziegler, and D. Scaramuzza, “Exploration without global consistency using local volume consolidation,” in *IFRR International Symposium on Robotics Research (ISRR)*, Hanoi, 2019, IFRR: IEEE, Oct. 2019. [Online]. Available: <https://doi.org/10.5167/uzh-197724>

Work experience

- 2026.09– **Postdoctoral Researcher**, *University of Toronto*, Toronto, Canada, 100%
 ○ Working on event-based state estimation in Tim Barfoot’s ASRL
 ○ Supervision of PhD, MSc and BSc students
 Technologies used: C++, Python, Julia, PyTorch, OpenCV, numpy, Eigen, ROS1/2, git, L^AT_EX
- 2021.06–2025.11 **PhD Candidate**, *University of Tübingen*, Tübingen, Germany, 100%
 ○ Working on event-based computer vision for fast robot control in collaboration with Sony AI contributing to project ACE
 ○ Supervision of MSc and BSc students
 ○ Teaching Assistant
 Technologies used: C++, Python, Julia, PyTorch, OpenCV, numpy, Eigen, ROS1/2, git, L^AT_EX
- 2023.11–2024.03 **Research Scientist Intern**, *Sony AI*, Zurich, Switzerland, 100%
 Worked on multi modal camera calibration in project ACE under the supervision of Dr. Raphaela Kreiser and Dr. Naoya Takahashi.
 Technologies used: C++, Python, OpenCV, Ceres, git
- 2022.08–2022.10 **Computer Vision & ML Research Intern**, *Prophesee*, Paris, France, 100%
 Worked on slow motion from frame and event data under the supervision of Dr. Amos Sironi.
 Technologies used: Python, PyTorch, OpenCV, numpy, git, Atlassian tools
- 2018.09–2021.05 **Robotics Engineer**, *MT-Robot AG*, Zwingen, Switzerland, 100%
 Accomplished tasks:
 ○ Development of a computer vision based safety field intrusion detection system
 ○ Improvement of a multi robot collision avoidance system
 ○ Development and maintenance of software for autonomous mobile robots (AMRs), including topics such as multi sensor fusion, mapping, path planning, (multi robot) obstacle avoidance, etc.
 ○ Deputy Scrum Master
 Technologies used: C++, Python, ROS1/2, DDS, OpenCV, CMake, git, Atlassian tools
- 2018.06–2018.09 **Research Assistant**, *University of Zürich, Robotics and Perception Group*, Zurich, Switzerland, 100%
 Continued working on my master thesis project which led to [13].
- 2018.04–2018.06 **Research Associate Intern**, *Disney Research Zürich*, Zürich, Switzerland, 100%
 Integrated a Leica total station in an existing ROS setup within the PaintCopter project.
 Technologies used: C++, Python, ROS, Ceres, CMake, git
- 2018.02–2018.03 **Research Assistant**, *Laboratory for Orthopaedic Biomechanics at the University and ETH Zürich*, Zürich, Switzerland, 100%
 Developed an LED light controller for a microscope setup which contributed to [12].
 Technologies used: C++, Qt, wxWidgets, CMake, git

- 2017.03–2017.08 **Computer Vision & Robotics Research Intern**, *Pix4D SA*, Lausanne, Switzerland, 100%
 Accomplished tasks:
 ○ Worked on indoor navigation for UAVs
 ○ Implementation of a filtering method for a robust target detection
 ○ Investigation of barcode localization and detection algorithms for automatic inventory
 Technologies used: C++, ROS, OpenCV, Eigen, Conan, CMake, Jenkins, git
- 2015.08–2018.06 **Software Engineer & System Administrator**, *Accelerom AG*, Zürich, Switzerland, 20%-30%
 Technologies used: Java, Groovy, JavaScript, jQuery, CSS, Grails, Hibernate, MySQL, git, Redmine, Tomcat, Apache, SAMBA
- 2014.02–2015.08 **Research Assistant**, *Laboratory for Orthopaedic Biomechanics at the University and ETH Zürich*, Zürich, Switzerland, 100%/20%
 Continued my work, provided further consulting and maintenance.
- 2013.11–2014.02 **Research Assistant (Civil service)**, *Computer Assisted Research and Development, University Hospital Balgrist*, Zürich, Switzerland, 100%
 Worked on segmentation algorithms for computer-assisted surgical planning.
 Technologies used: Matlab, C#, VTK, CVS
- 2013.08–2013.11 **Research Assistant (Civil service)**, *Laboratory for Orthopaedic Biomechanics at the University and ETH Zürich*, Zürich, Switzerland, 100%
 Accomplished tasks:
 ○ Extended and adapted a microscope control software
 ○ Developed and implemented a stretcher control software
 Technologies used: C++, Qt, wxWidgets, CMake, git
- 2008.08–2009.03 **Computer Science Intern**, *ERPsourcing AG*, Wallisellen, Switzerland, 100%
- 2004.08–2008.08 **Electronics Engineer Apprentice**, *Hch. Kündig & Cie. AG*, Rüti ZH, Switzerland, 100%

Independent coursework and training

- Startup Center
Tübingen Leadership Talent Academy. Certificate earned on November, 2024
- SIY Search Inside Yourself: Emotional Intelligence for Leadership. Certificate earned on May 28, 2024
- Center for
Nonviolent
Communication NVC Courses and Workshops. Certificates earned in 2024 and 2025
- University of
Tübingen Career Development through Research Funding. Certificate earned on October 7, 2025
- University of
Tübingen Writing successful grant proposals. Certificate earned on July 11, 2025
- University of
Tübingen From Chaos to Structure: Time- and Self-Management for a More Efficient Work Style. Certificate earned on March 9, 2022
- University of
Tübingen Slidewriting and Storylining. Certificate earned on May 16, 2022
- University of
Tübingen Setting and Achieving Goals Efficiently. Certificate earned on May 5, 2022

- University of Tübingen Speedreading. Certificate earned on April 14, 2022
- edX DT-01x: Self-Driving Cars with Duckietown by ETHx on edX. Specialization Certificate earned on August 15, 2021
- Coursera Deep Learning, a 5-course specialization by deeplearning.ai on Coursera. Specialization Certificate earned on March 16, 2018
- edX Autonomous Mobile Robots by ETHx on edX. Certificate earned on April 17, 2014

Awards and Media coverage

- The Guardian AI-powered robot beats elite table tennis players
- Schwäbisches Tagblatt Forscherteam der Uni Tübingen entwickelt Tischtennis-Roboter
- University of Tübingen Scholarship for the Leadership Talent Academy

Academic services

- Conference workshops Workshop on Neuromorphic Perception for Real World Robotics (NeuRobots), IROS 2025
- Workshop on Neuromorphic Robotic Systems (NeuRoSys), RSS 2026
- Paper reviews IROS, ICRA, CoRL, CVPR, ECCV, ACCV, RA-L, T-ASE, T-RO

Supervised thesis

- Bachelor thesis
 - Simulating event-based cameras with frame-based cameras, Daniel Teigland
 - Deep-learning based table tennis ball tracking with an event camera, Genc Ahmeti
 - Event-camera, camera and robot arm calibration, Levin Kolmar
 - 3D trajectory prediction from event data, Julian John
 - Pushing an event-simulator towards its limit, Laura Schiller
 - Adding noise and artifacts to the event-simulator, Steven Krämer
 - Event-based camera bias optimization, Eric Langlouis
 - Multi object tracking via event-based motion segmentation with event cameras, Zhiyu Han
 - Automatic bias optimization for event cameras using Reinforcement Learning, Emil Moldovan
 - Automatic bias optimization for event cameras using offline Reinforcement Learning, David Joseph
 - Grasping a table tennis ball using Model Predictive Control, David Grawe
 - Asynchronous Graph-based Neural Networks for Ball Detection with Event Cameras, Adrian Bytyqi
 - Spin estimation of a table tennis ball with an event-camera, Maximilian Schnitt
 - Event-Based vision for a tactile sensor, Alexander Löwe
 - Ball Detection with event-based asynchronous sparse convolutional networks, Emre Sözbilir

Asynchronous Graph-based Neural Networks for Ball Detection with Event Cameras, Sonja Baumann

Master thesis Spiking neural network for event based ball detection, Karl Vetter

Event-based spin estimation, Julian Krismer

Real-time MPC control of an industrial robot arm for table tennis, Till Köpf

Automatic bias optimization for event cameras using Inverse Reinforcement Learning, Manuel Rauschenberger

Learning-based spin estimation of table tennis ball with an event-camera, Levin Kolmar

Teaching activities

SS 2025 Teaching Assistant: Mobile Robots

AS 2024 Teaching Assistant: Foundations of Robotics

SS 2024 Teaching Assistant: Mobile Robots

AS 2023 Seminar: Robotics and Robot Vision

AS 2022 Teaching Assistant: Introduction to Computer Engineering

SS 2022 Teaching Assistant: Mobile Robots

AS 2021 Teaching Assistant: Deep Learning

Languages

German Mother tongue

English Excellent, Level C1

French Good, Level B1

Korean Basics, Level A2

Chinese Basics, Level A1

Technical skills

Languages C++, Python, Julia, C, Java

Software packages OpenCV, ROS1/2, PyTorch, Eigen, boost, DDS

Hobbies

Sports Wing Chun Kung Fu, Yoga, Jogging, Mountaineering

Music Drums, Piano, Vocals

Extra-Curricular activities

- Co-clerk at Switzerland Yearly Meeting (Quakers)
- Member of the Executive Committee at Quaker Council for European Affairs (QCEA)